



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI NAVAMANI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

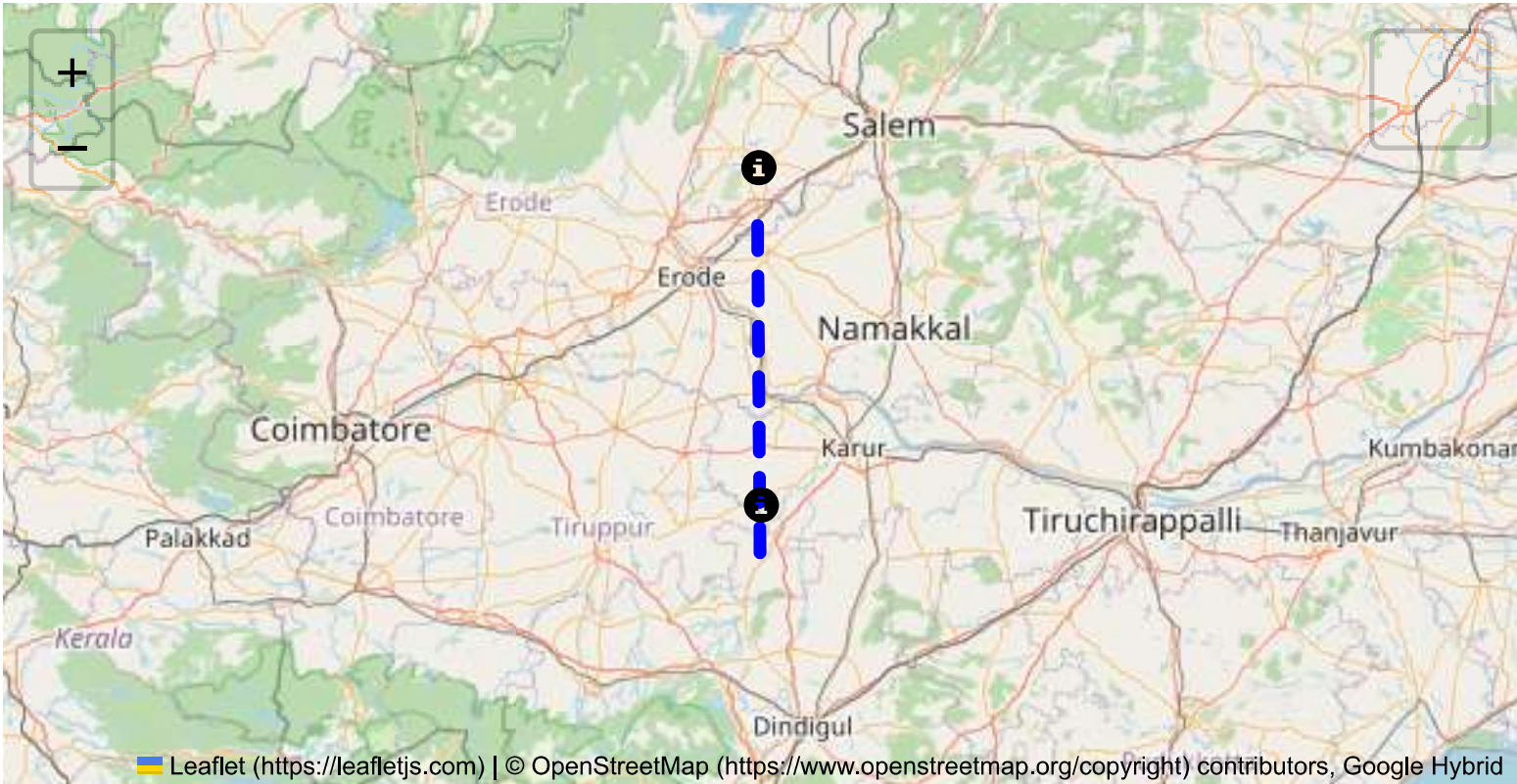
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

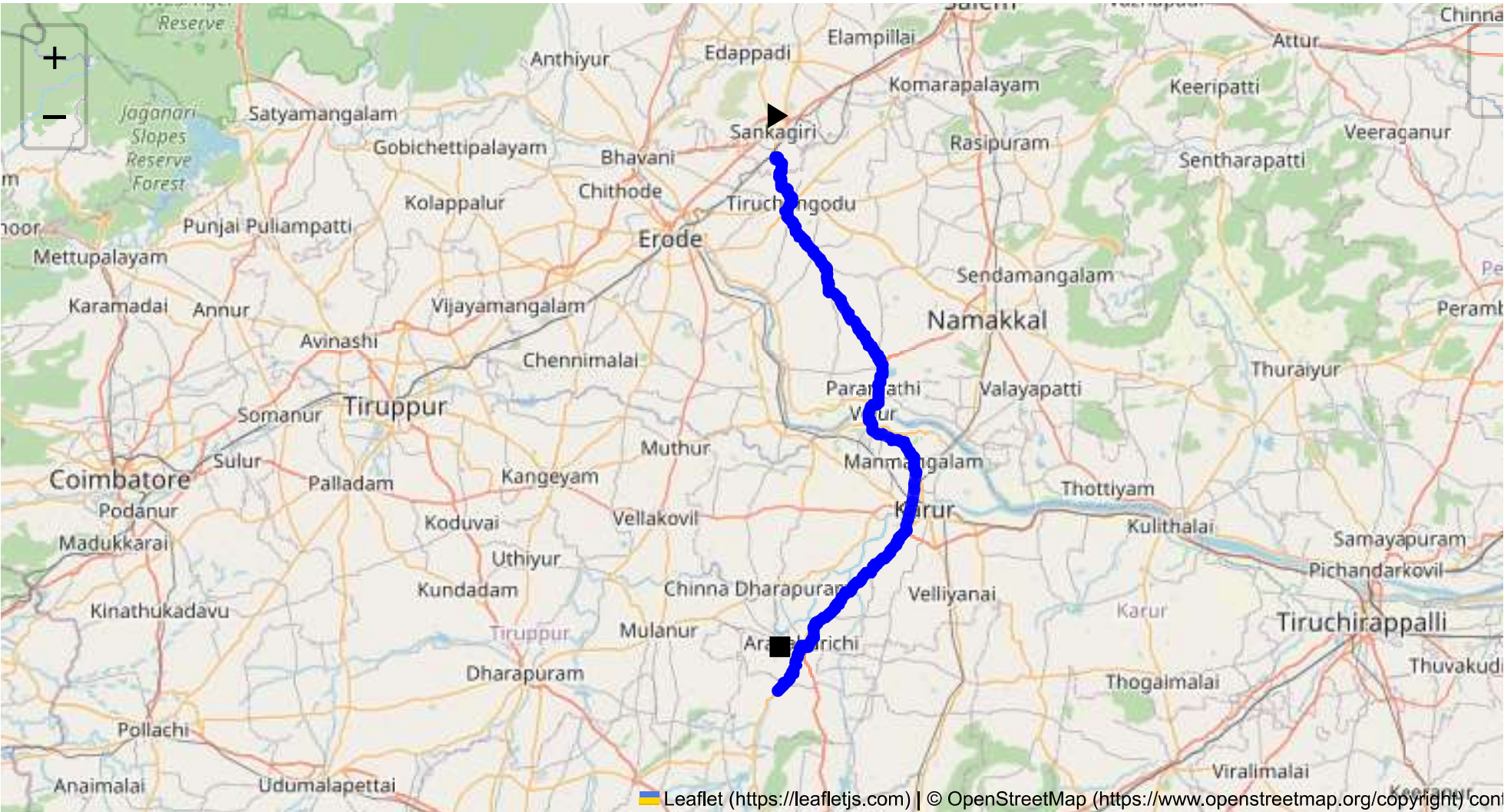
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 100.94 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (10.712294, 77.87724)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Pallapatti in Tamil Nadu spans approximately 100.94 kilometers. The journey typically follows National and State Highways, passing through a mix of urban and rural landscapes. Key waypoints include Salem, Namakkal, and Karur before reaching Pallapatti.

Weather Conditions

- 1. **Typical Weather:** The region experiences a tropical climate with hot summers (March-May) and moderate monsoons (June-September).
- 2. **Potential Hazards:** Monsoon rains can lead to waterlogging and reduced visibility. Heat during summer months may affect vehicle performance.

Traffic Patterns

- 1. **Peak Hours:** Traffic tends to be heavier during morning (8-10 am) and evening (6-8 pm) hours.
- 2. **Congestion-Prone Areas:** Urban zones near Salem and Karur tend to experience congestion.

Road Quality and Infrastructure

- 1. **Road Quality:** Generally good on National Highways but can deteriorate on local roads near Pallapatti.
- 2. **Infrastructure:** Adequate signage and lighting are present on major highways, but rural segments may lack both.

Alternative Routes

- 1. **Emergency Detours:** Secondary routes via SH-34 and SH-95 can be considered as detours around congested areas or roadblocks.

Regulations on Hazardous Material Transport

- 1. **Local Regulations:** Must comply with TN Motor Vehicles Act and Central Motor Vehicles Act, requiring valid permits for hazardous material.

Historical Incidents

- 1. **Reports of Accidents:** The route has seen occasional accidents, mostly attributed to driver fatigue and adverse weather conditions.

Environmental Considerations

- 1. **Sensitive Areas:** Rivers and agricultural lands near Karur require caution to avoid contamination during spillage incidents.

Communication Coverage

- 1. **Dead Zones:** Spotty coverage might occur in rural segments, notably as the route approaches Pallapatti.

Emergency Response Times

- 1. Estimation:** Urban centers such as Salem and Karur have quicker response times (~20-30 minutes) compared to more remote areas where it can extend beyond an hour.

Overall Summary

Despite being a major route, the journey between Sangagiri and Pallapatti involves challenges like potential weather impacts and some communication issues. Drivers must maintain vigilance particularly in rural areas with less infrastructure support. Thorough knowledge of regulations, alongside contingency awareness for weather and traffic conditions, can mitigate most risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	High	11.37862, 77.89488	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16764, 78.01565	30 KM/Hr
7	Turn	High	11.14545, 78.02208	15 KM/Hr
8	Turn	High	11.14480, 78.02291	15 KM/Hr
9	Turn	Medium	11.10102, 78.00572	30 KM/Hr
10	Turn	Medium	10.89913, 78.03032	30 KM/Hr
11	Turn	Medium	10.79857, 77.92714	30 KM/Hr
12	Turn	Medium	10.77148, 77.91668	30 KM/Hr
13	Turn	Medium	10.77140, 77.91639	30 KM/Hr
14	Turn	Medium	10.77216, 77.91277	30 KM/Hr
15	Turn	High	10.77350, 77.90980	15 KM/Hr
16	Turn	Medium	10.72943, 77.89224	30 KM/Hr
17	Turn	High	10.72608, 77.89166	15 KM/Hr
18	Turn	High	10.71243, 77.87776	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
19	hospital	Government Hospital, Aravakurtchi	10.77208, 77.909648	30 km/h	Medium
20	hospital	Pallapatti Government Hospital	10.72384, 77.891681	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10102, 78.00572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89913, 78.03032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79857, 77.92714



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.77148, 77.91668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.77216, 77.91277



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.77350, 77.90980



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.72943, 77.89224



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.72608, 77.89166



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.71243, 77.87776

